

Senior Data Scientist & Software Developer at Urban Analytics startup out of Harvard Innovation Labs. Specialising in digital twins, advanced ML, econometrics and complex systems analysis | *MSc Geospatial Data Science, MA Economics*

EDUCATION

M.Sc Spatial Data Science | University College London | *Distinction* **2023 – 2024**

- Specialised in applying complexity science and ML to geospatial problems at *The Bartlett School of Built Environment*
- Awarded funding (£2000) from Space Syntax Ltd. and the Consumer Data Research Centre to write MSc dissertation

M.A (Hons) Economics | University of Edinburgh | *First Class* **2016 – 2020**

- **Dissertation:** *The impact of urban sprawl and modern zoning laws on indigenous communities in Western Canada*

International Baccalaureate | United World College Maastricht | *Summa Cum Laude 40/45* **2013 – 2015**

- Awarded 2 year Academic Scholarship: £40 000 | 96th percentile internationally

WORK EXPERIENCE

University of Oxford | Inst. for New Economic Thinking | Volunteer Researcher | Oxford, UK **Apr 2026 – Present**

- Contributing technology cost data & analysis to Oxford Martin School's Programme on *Forecasting Technological Change* w/ Prof. Francis Lafond & Dr. Amir Akther. To be credited in public database on clean tech learning curves
- Supporting research that tracks how clean technologies improve & become cheaper over time helping inform UK climate policy & investment decisions
- Conducted sensitivity analysis to test the robustness of cost forecasts under different assumptions & scenarios
- Developed & validated learning curve models to forecast future cost reductions of climate critical UK technologies

Aretian Urban Analytics | Senior Data Scientist & SWE | London, UK & Cambridge, USA **Nov 2025 – Present**

- Using causal inference, systems analytics and geospatial ML to inform economic strategy for cities & regions
- *Large scale web-scraping:* Developed automated pipeline to extract and harmonise financial and project data from unstructured PDF reports, linking to regional economic indicators for cross-city comparison
- Building data pipelines in AWS for city-scale digital twins, combining 100+ variables into urban performance model
- Spatial data and network visualisation for a variety of urban analytics and econometric projects in AWS, MSFT suite
- Co-authored research paper which models the innovation ecosystem of the Boston Cambridge Somerville Region
- Applying Economic Complexity methods (PCI and PDI) to advise on how cities can develop more resilient economies
- Measured urban form & street network structure to assess how physical layout relates to economic performance
- Applying complex geospatial ML methods to assess resilience of street & transportation networks for various cities

Norman Foster Institute | Data Scientist | Madrid, Spain **Jan 2025 – Dec 2025**

- Awarded **€52 000** grant to research Spatial Data Science & Sustainable Development in Cape Town, South Africa
- Developing *Densification Potential Model* to identify areas of unrealised economic potential in Cape Town. Tool built in Google Earth Engine using LIDAR, housing capacity, population density, Land Use Zoning in Python & JavaScript
- Developed Cape Town *Public Space Qualification Tool* to assess their quality in line with municipal development plan

Space Syntax Ltd. | Data Science Researcher | London, UK **Apr 2024 – Sept 2024**

- **Dissertation Title:** *Using unsupervised machine learning methods to identify multiscale centrality indicators*
- Authored research paper in conjunction with Space Syntax Ltd. which predicts the location of economic activity based on the mathematical layout of UK street networks and how this model changes at different scales
- Examined the layout of transportation & street networks to predict optimal business locations, leading to improved economic activity, promotion of sustainable transport and urban development practices

- Offered full time Spatial Data Scientist role at Space Syntax Ltd. based on skillset and quality of work

Baringa Partners | Management Consultant (Software & Data) | London, UK

Mar 2021 – Aug 2023

Designed organisational structure of Vodafone consumer division for future merger. Delivered ~ £10m OpEx reduction

- Spearheaded design of Data & Analytics team. Collaborated closely w/ senior leadership through iterative workshops
- Developed 6-month implementation & communication plan to ensure seamless transition to new organisation structure

Developed technology & AI strategy for *Magic Circle* law firm Freshfields Bruckhaus Deringer

- Designed 2 data pipelines, which ingest legal data and provide insights on the likelihood of winning a dispute, legal costs, likely pay-out amounts and overarching legal trends in certain industries

Designed data architecture which helped Thomson Reuters develop a more targeted marketing and sales strategy

- Used data visualisation software (PowerBI) to represent client's sales performance and profitability over time

Agile Project Manager: Developed data pipeline for largest music licensing company in the UK (PPL PRS)

- Built data pipeline to increase data ingestion speed by 500% and produce insights from unutilised metadata

Strategist working on the largest supply chain transformation in UK Publishing history (Penguin Random House)

- Designed supply chain of UK's largest book publisher to accommodate COVID demand spike & a recent acquisition
- Developed & led 10+ training sessions to ensure new analytics software was correctly utilised: staff now had fewer touchpoints, more storage recommendations by SKU & new conveyer belts

Q5 Partners | Management Consultant (Intern) | London, UK

Sept 2020 – Mar 2021

Worked on a restructuring and organisational design project for National Grid (£14b revenue utilities company)

- Used Geographic Information Systems to expand National Grid's electricity grid in West England

Boston Consulting Group | Mobility Analyst | London, UK

Sept 2018 – Oct 2019

- Student researcher at the BCG where I focussed on Autonomous Vehicles, Smart Cities & Intermodal Applications

ACHIEVEMENTS

- €52 000 Research grant from the Norman Foster Institute to study Sustainable Cities and Data Science | 01/2025
- 3rd Place | Dissertation awarded 3rd at the Society for Location Analysis *UK Masters Dissertation Awards* | 11/2024
- 1st Place | Centre for Advanced Spatial Analysis Hackathon w/ West Midlands Fire Service | £2000 Cash Prize | 06/2024
- 1st Place | Boston Consulting Group *London University Case Study Competition* | £500 Cash Prize | 09/2019

TECHNICAL SKILLS & INTERESTS

Language & Libraries | *R*, *Javascript*, *SQL*, *Python* (sklearn, spaCy, NumPy, SciPy, Matplotlib, Seaborn, NetworkX)

Software | AWS, GCP, Google Earth Engine, GitHub, QGIS, CARTO, Snowflake, MS Office Suite, Gephi

Areas of Practice | Causal Inference, Neural Networks, Remote Sensing, Climate Modelling, Urban Simulation, Data Architecture, Machine Learning, Data Visualisation, Project Management, Stakeholder Management, Supply Chain & Procurement, Operating Model

Interests | Climbing, Salmon fishing, Alpinism, Avid reader, Film, Chess, Diversity Equity & Inclusion

[GitHub Portfolio](#)
